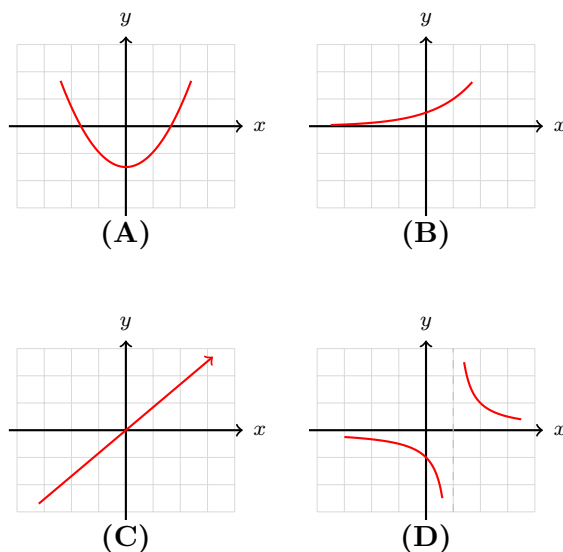


Lesson 4.3: Multiply and Divide Rational Expressions

Math Objectives	Identify the values that must be excluded from the domain of a rational expression. Simplify rational expressions by factoring and canceling common factors, then state the domain. Build and interpret rational expressions that arise from real-world ratios (geometry, probability, efficiency).
Language Objective	Students will explain why a domain restriction must come from the <i>original</i> denominator (not the simplified one), verbally and in writing, with the support of structured talk-time, models, and real-time scaffolding/questioning.
Essential Question	What stays the same and what changes when we rewrite a rational expression in simplified form?

Do Now

Part 1. Match each graph (A, B, C, D) to its function family.



Word bank: linear • quadratic • exponential • rational

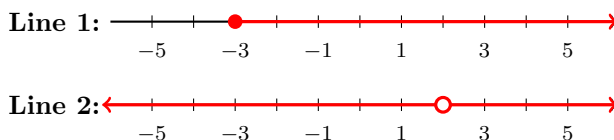
(A) is _____

(B) is _____

(C) is _____

(D) is _____

Part 2. Two number lines. What's different about them?



What does the closed (filled-in) circle on Line 1 mean?

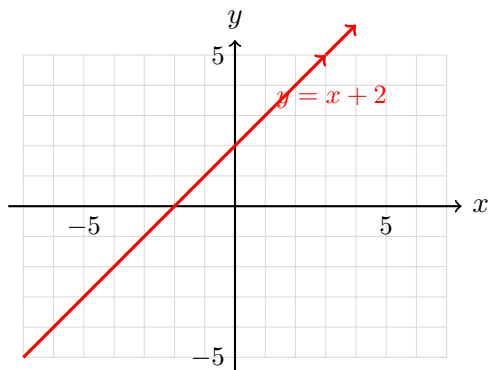
What does the open (hollow) circle on Line 2 mean?

Write each as a domain expression: Line 1: _____

Line 2: _____

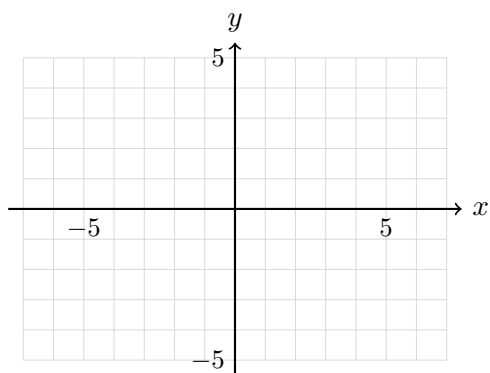
Explore & Reason

Consider the following graph of the function $y = x + 2$.



A. What is the domain of this function?

B. Sketch a function that resembles the graph, but *restrict its domain to exclude $x = 2$* .

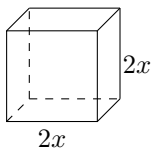


C. **Use Structure.** What *kind of function* might have a graph like the one you sketched in part B? Explain.

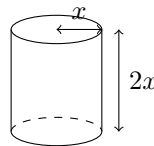
Frame: “The graph could come from a function whose denominator is _____, because that makes $x =$ _____ excluded.”

Example 6 – Cube vs Cylinder (guided)

A company is comparing two package designs. The more efficient design has the **smaller** ratio of surface area to volume (SA/V).



Option 1: Cube



Option 2: Cylinder

We will work the **cube** together. Then you will compute the **cylinder** on your own.

Option 1 – Cube (we do)

$$V = (2x)^3 = 8x^3 \quad SA = 6(2x)^2 = 24x^2$$

$$\frac{SA}{V} = \frac{24x^2}{8x^3} = \frac{3}{x}$$

Option 2 – Cylinder (your turn)

Use $V = \pi r^2 h$ and $SA = 2\pi r h + 2\pi r^2$, with $r = x$ and $h = 2x$.

$$V = \underline{\hspace{4cm}}$$

$$SA = \underline{\hspace{4cm}}$$

$$\frac{SA}{V} = \underline{\hspace{4cm}}$$

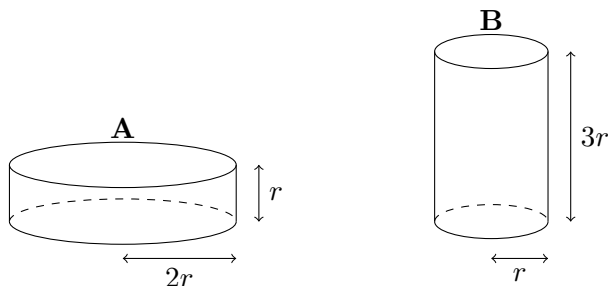
Comparison. Both ratios should come out to $\frac{3}{x}$. What does that tell you about the design choice?

Frame: “The ratios are $\frac{3}{x}$, so the company should choose based on $\frac{3}{x}$ (cost, manufacturing, etc.)”

Apply – Make Sense and Persevere

An engineering firm wants to construct a cylindrical structure that will *maximize the volume for a given surface area*. Compare the ratios of the volume to surface area of each of the cylindrical structures shown, using the following formulas for volume and surface area of cylinders.

$$V = \pi r^2 h \quad SA = 2\pi r h + 2\pi r^2$$



a. Calculate the ratio of volume to surface area for cylinder A.

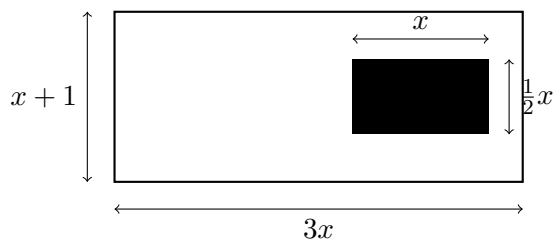
b. Calculate the ratio of volume to surface area for cylinder B.

c. Which of these cylinders has a greater ratio of volume to surface area? *What does that mean about the design?*

Frame: “Cylinder _____ has the greater ratio because _____, which means a designer who wants the most volume per unit of material should pick _____.”

Performance Task

Lupe designed a carnival game that involves tossing a beanbag into the box shown. To win a prize, the beanbag must fall inside the **black rectangle**. The probability of winning is equal to the ratio of the area of the black rectangle to the total area of the face of the box shown.



A. Write the rational expression for the probability of winning. *Show what you used for the numerator and denominator.*

B. Simplify the expression. State the domain restrictions and explain what they mean *in this physical situation* (the carnival box).

Frame: "The simplified expression is _____. The value $x =$ _____ is excluded because _____."

C. Suppose Lupe builds the game with $x = 4$ feet. What is the probability that a randomly thrown beanbag wins? *Would you say this is a fair carnival game? Why or why not?*

Frame: "When $x = 4$, the probability is _____. I think the game is _____ because _____."

Name _____

Exit Ticket

Use Appropriate Tools. Explain why domain restrictions are necessary to show that the rational expressions $\frac{-6x^2 + 21x}{3x}$ and $-2x + 7$ are equivalent. What is true about the graphs at $x = 0$, and why?

You may use any of these sentence frames if they help:

- “At $x =$ _____, the original expression $\frac{-6x^2 + 21x}{3x}$ is undefined because _____.”
- “The graph of $\frac{-6x^2 + 21x}{3x}$ has a _____ at $x = 0$, but the graph of $-2x + 7$ does not.”
- “Without the domain restriction, the two expressions are not _____.”

What is something you learned today?

What is something you liked about this lesson?